1.9.1 Powers with positive exponents

Value of powers given by positive whole number exponents quite often increase rapidly. Observe the following example:

$2^1 = 2$

$2^2 = 2 \times 2 = 4$

$2^3 = 2 \times 2 \times 2 = 8$

$2^4 = 2 \times 2 \times 2 \times 2 = 16$

$2^5 = 2 \times 2 \times 2 \times 2 \times 2 = 32$

$2^6 = 2 \times 2 \times 2 \times 2 \times 2 \times 2 = 64$

$2^7 = 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 = 128$

$2^8 = 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 = 256$

$2^9 = 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 = 512$

$2^{10} = 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 \times 2 = 1024$

At this rate of increase, what do you think 2^{100} will be?

In fact, **$2^{100} = 1267650600228229401496703205376$**

Thus, we understand that the positive exponential notation with positive power could be useful when we come across with large numbers.

1.9.2 Powers with zero and negative exponents

Observe this pattern:

$2^5 = 32$

$2^4 = 16$

$2^3 = 8$

$2^2 = 4$

$2^1 = 2$

$2^0 = ?$

Starting from the beginning, what happens in the successive steps? We find that the result is half of that of the previous step. So, what can we say about 2^0 ? If we prepare a table like this for 3^5, 3^4, 3^3, and so on what will it tell us about 3^0? We can use the same process as in this pattern, to conclude that any non-zero number raised to the zero exponent must result in 1. Thus,

$a^0 = 1$, where **$a \neq 0$**

Let us see what happens if we extend the above pattern further downward.

As before, starting from the beginning, in the successive steps, we find that the result is half of that of the previous step. Since $2^0 = 1$, the next step is 2^{-1}, whose value is the previous step's value 1, divided by 2, that is $\frac{1}{2}$. Next is 2^{-2}, which is the same as $\frac{1}{2}$ divided by 2, that is $\frac{1}{4}$ and so on. Thus,

$2^3 = 8$

$2^2 = 4$

$2^1 = 2$

$2^0 = 1$

$2^{-1} = \frac{1}{2}$

$2^{-2} = \frac{1}{4}$

$2^{-3} = \frac{1}{8}$

In general, $a^{-m} = \frac{1}{a^m}$, where m is an integer

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